

formulas important.

Cheatsheet for Reinforcement Learning

value function

$$V^\pi(s) = \mathbb{E} \left(\sum_{t=0}^{\infty} \gamma^t r_t \mid s_1 = s \right) \text{ finite}$$

$$V^\pi(s) = \mathbb{E} \left(\sum_{t=0}^{\infty} \gamma^t r_t \mid s_1 = s \right) \text{ infinite}$$

Bellman equations

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(s), s' \sim P(s,a)} [R(s,a,s') + \gamma V^\pi(s')] \text{ finite horizon}$$

Optimal value and optimal policy

$$V^*(s) = \sup_{\pi} V^\pi(s) = \max_a [R(s,a) + \gamma \sum_{s'} P(s,a,s') V^*(s')] \text{ u.p.l.b.}$$

$$= \max_a [R(s,a) + \gamma \mathbb{E}_{s' \sim P(s,a)} [V^*(s')]]$$

$$\pi^*(s) = \arg \max_a [R(s,a) + \gamma \mathbb{E}_{s' \sim P(s,a)} [V^*(s')]]$$

Q-value and its Bellman equation

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s,a,s') V^*(s')$$

$$= R(s,a) + \gamma \sum_{s'} P(s,a,s') (\max_{a'} Q^*(s,a'))$$

$$Q^\pi(s,a) = R(s,a) + \gamma \sum_{s'} P(s,a,s') V^\pi(s')$$

$$= R(s,a) + \gamma \sum_{s'} P(s,a,s') \mathbb{E}_{a' \sim \pi(s)} [Q^\pi(s',a')]$$

Advantage function

$$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$$

→ optimal policy $V^* = \max_a Q^*(s,a)$

$$\pi^*(s) = \arg \max_a Q^*(s,a)$$

It can be any policy calculated by Q-values: $\pi(s) = \arg \max_a Q(s,a)$

$$\text{sigmoid function: } \sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

$$\sigma'(\theta) = \sigma(\theta)(1 - \sigma(\theta))$$

$$V^\pi(s) = \sum_a \pi(s,a) Q^\pi(s,a)$$

也可用来计算 value (V 是 Q 的 convex combination function)

$$R(s,a) = \sum_{s'} P(s,a,s') R(s,a,s')$$

average reward

$$\rho^\pi(s) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T r_t \mid s_1 = s, a_t = \pi(s_t) \right]$$

bias of a policy

$$h^\pi(s) = \lim_{T \rightarrow \infty} \mathbb{E} \left[\sum_{t=1}^T (r_t - \rho^\pi(s)) \mid s_1 = s, a_t = \pi(s_t) \right]$$

Q-Learning

1. initialize $Q \in \mathbb{R}^{S \times A}$

2. in every iteration $k=0,1,2,\dots$

$$Q^k(s,a) \leftarrow R(s,a) + \gamma \sum_{s'} P(s,a,s') (\max_{a'} Q^k(s',a'))$$

3. stop if $\|Q^k - Q^{k-1}\|$ is small

4. optimal policy $\pi(s) = \arg \max_a Q^k(s,a)$

For Q-learning in every iteration

→ observe s_k , take action

$$a_k = \arg \max_a Q^k(s_k, a) \text{ u.p.l.b.}$$

→ improve $Q(s_k, a_k)$ as

$$Q^k(s_k, a_k) \leftarrow (1 - \alpha_k) Q^k(s_k, a_k) + \alpha_k (r_k + \gamma \max_{a'} Q^k(s_{k+1}, a'))$$

$$\text{i.e. } Q^k(s_k, a_k) \leftarrow (Q^k(s_k, a_k) + \alpha_k \delta_k)$$

$$\text{TD: } \delta_k = r_k + \gamma \max_{a'} Q^k(s_{k+1}, a') - Q^k(s_k, a_k)$$

Large scale Q-learning $Q \rightarrow Q_\theta$

$$Q_k \leftarrow Q_{k-1} - \alpha_k \nabla_{\theta} L(Q_{k-1})$$

$$L(Q_{k-1}) = L(Q_{k-1}(s_k, a_k), r_k + \gamma \max_{a'} Q_{k-1}(s_{k+1}, a'))$$

if using squared loss

$$Q_k \leftarrow Q_{k-1} + \alpha_k \delta_k \nabla_{\theta} Q_{k-1}(s, a)$$

where TD target: $y = r_k + \gamma \max_{a'} Q_{k-1}(s_{k+1}, a')$

$$\text{TD error: } \delta_k = r_k + y - Q_{k-1}(s_k, a_k)$$

$$V^\pi(s) = \sum_a \pi(s,a) [R(s,a) + \gamma \sum_{s'} P(s,a,s') V^\pi(s')]$$

Q-value iteration

Policy gradient methods

parameterize the policy as

$$\pi_\theta(s,a) = \frac{e^{\theta_a(s)}}{\sum_{a' \in A} e^{\theta_{a'}(s)}}$$

$$\pi_\theta(s) = \mathcal{N}(\theta(s), \Sigma)$$

$$\text{goal: } \theta^* = \arg \max_{\theta} V^{\pi_\theta}(s)$$

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k \nabla_{\theta} V^{\pi_\theta}(s)$$

Finite horizon

$$\nabla_{\theta} V^{\pi_\theta}(s) = \mathbb{E}_T \left[R(T) \sum_{t=1}^H \nabla_{\theta} \log(\pi_\theta(s_t, a_t)) \mid s_1 \right]$$

in every iteration

→ execute current policy to obtain sample trajectories $\tau_i, i=1,2,\dots,m$

$$R^i(T) = r_1 + \gamma r_2 + \dots + \gamma^{H-1} r_H$$

$$\text{calculation } \bar{R}(T_i) \text{ for } i=1,2,\dots,m$$

$$\bar{g}_k = \frac{1}{m} \sum_{i=1}^m \bar{R}(T_i) \sum_{t=1}^H \nabla_{\theta} (\log(\pi_\theta(s_t, a_t)))$$

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k \bar{g}_k$$

Infinite horizon

$$V^{\pi_\theta}(s) = \lim_{T \rightarrow \infty} \mathbb{E} \left(\sum_{t=1}^T \gamma^{t-1} r_t \mid s_1 \right)$$

$$= \sum_a d^{\pi_\theta}(s \mid s_1) \sum_a \pi_\theta(s,a) R(s,a)$$

$$d^{\pi_\theta}(s \mid s_1) = \lim_{T \rightarrow \infty} \sum_{t=1}^T \gamma^{t-1} P(s_t = s \mid s_1, \pi)$$

$$\sum_s (1 - \gamma) d^{\pi_\theta}(s) = 1$$

Theorem (里面 α 可以用 A 替换)

$$\nabla_{\theta} V^{\pi_\theta}(s) = \sum_s d^{\pi_\theta}(s \mid s_1) \sum_a \pi_\theta(s,a) \nabla_{\theta} Q^{\pi_\theta}(s,a)$$

$$= \sum_s d^{\pi_\theta}(s \mid s_1) \sum_a \pi_\theta(s,a) Q^{\pi_\theta}(s,a) \nabla_{\theta} \log(\pi_\theta(s,a))$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \gamma d^{\pi_\theta}} [Q^{\pi_\theta}(s,a) \nabla_{\theta} \log(\pi_\theta(s,a))]$$

Policy gradient in every iteration:

→ execute current policy to obtain trajectories

→ for any trajectory, compute

$$Q_t^i = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$$

$$\bar{g}_k = \frac{1}{m} \sum_{i=1}^m \sum_{t=1}^T \gamma^{t-1} Q_t^i \nabla_{\theta} \log(\pi_\theta(s_t, a_t^i))$$

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k \bar{g}_k$$

Tip: 如果有 absorbing state, $A(s,a)=0$

$$\therefore \nabla_{\theta} V^{\pi_\theta}(s) \propto \sum_a \pi_\theta(s,a) A^\pi(s,a) \nabla_{\theta} \log(\pi_\theta(s,a))$$

只计算不是 absorbing 的状态

Performance Difference Lemma

For any $\tilde{\pi}, \pi$ start at same s_1

$$V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1) = \sum_s d^{\tilde{\pi}}(s) \sum_a \tilde{\pi}(s,a) A^{\pi}(s,a)$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim (1-\gamma) d^{\tilde{\pi}}, a \sim \tilde{\pi}} [A^{\pi}(s,a)]$$

or

$$V^{\pi'}(s_1) - V^{\pi}(s_1) = \sum_s d^{\pi'}(s) \sum_a \pi'(s,a) A^{\pi}(s,a)$$

Policy improvement theorem

For any π, π' , let $\pi^{\text{new}} = (1-\alpha)\pi + \alpha\pi'$

then we have

$$V^{\pi^{\text{new}}}(s_1) - V^{\pi}(s_1) \geq \frac{\alpha}{1-\gamma} A_{\pi}(\pi') - \frac{\alpha^2}{(1-\gamma)^2} \gamma A_{\pi}^{\text{max}}$$

$$A_{\pi}(\pi') = \mathbb{E}_{s \sim (1-\gamma) d^{\pi}, a \sim \pi'} \left[\sum_a \pi'(s,a) A^{\pi}(s,a) \right]$$

$$A_{\pi}^{\text{max}} = \max_s \left| \sum_a \pi'(s,a) A^{\pi}(s,a) \right|$$

how to design an always good improvement

$$\rightarrow \alpha = \frac{A_{\pi}(\pi') (1-\gamma)^2}{4R} \quad R \text{ is upper bound reward}$$

$$A_{\pi}(\pi') \leq A_{\pi}^{\text{max}} \leq \frac{R}{1-\gamma}$$

$$\rightarrow \pi'(s) = \arg \max_a A^{\pi}(s,a)$$

conservative greedy policy improvement

$$\pi^{k+1} \leftarrow (1-\alpha)\pi^k + \alpha \pi^{k+1}$$

$$\pi^{k+1}(s) = \arg \max_a [Q^{\pi^k}(s,a) - V^{\pi^k}(s)]$$

$$\text{or } \pi^{k+1} \leftarrow (1-\alpha)\pi^k + \alpha \pi_{\theta^{k+1}}$$

$$Q^{\pi^k}(s) = \arg \max_{\theta} \mathbb{E}_{s \sim (1-\gamma) d^{\pi^k}, a \sim \pi_{\theta}} \left[\sum_a \pi_{\theta}(s,a) A^{\pi^k}(s,a) \right]$$

$$V(\pi^{\text{new}}) - V(\pi) \geq \frac{\alpha}{1-\gamma} A_{\pi}(\pi') - \frac{\alpha^2 \gamma}{(1-\gamma)^2} \frac{A_{\pi}^{\text{max}}}{8R}$$

algorithm:

1. initialize π

2. Repeat

→ (p1) play π from starting state dist μ to get trajectories. estimate $A^{\pi}(s,a)$ by mc and

→ (p2) compute π' as

$$\max_{\pi' \in \Pi} \mathbb{E}_{s \sim \gamma d^{\pi}, a \sim \pi'} \left[\sum_a \pi'(s,a) A^{\pi}(s,a) \right] = A'$$

$$\alpha = A' \frac{(1-\gamma)^2}{4R}$$

$$\text{update } \pi \leftarrow (1-\alpha)\pi + \alpha\pi'$$

$$\text{(TRPO)} \quad \theta_{k+1} = \arg \max_{\theta} \mathbb{E}_{s \sim \pi_{\theta}} \left[\sum_a \pi_{\theta}(s,a) A^{\pi_{\theta}}(s,a) \right]$$

$$\text{s.t. } \mathbb{E}_{s \sim \pi_{\theta}} [D_{KL}(\pi_{\theta}(\cdot|s), \pi_k(\cdot|s))] \leq \delta$$

$$\text{(DP0)} \quad r_i(\theta) = \frac{\pi_{\theta}(s_i, a_i)}{\pi_{\theta_k}(s_i, a_i)}$$

$$\theta_{k+1} = \max_{\theta} \mathbb{E}_i [\min(h_i(\theta) \bar{A}_i, \text{clip}(h_i(\theta), 1-\epsilon, 1+\epsilon) \bar{A}_i)]$$

softmax 的梯度计算问题. $\theta = (\theta_1, \theta_2)$

$$\pi_\theta(s, a) = \frac{\exp(\theta^T \phi_{s,a})}{\exp(\theta^T \phi_{s,a_1}) + \exp(\theta^T \phi_{s,a_2})}$$

$$\phi(s_1, a_1) = (1, 0) \quad \phi(s_1, a_2) = (0, 1) \quad \phi(s_2, a_1) = (0, 1) \quad \phi(s_2, a_2) = (1, 0)$$

$$\text{let } \Delta = \theta_1 - \theta_2 \quad \Delta(\Delta) = \frac{1}{1 + e^{-\Delta}}$$

$$\pi_\theta(s_1, a_1) = \Delta(\Delta) \quad \pi_\theta(s_1, a_2) = 1 - \Delta(\Delta) = \pi_\theta(s_2, a_1)$$

$$\log \pi_\theta(s, a) = \theta^T \phi_{s,a} - \log(\sum_{a'} \exp(\theta^T \phi_{s,a'}))$$

$$\nabla_{\theta} \log \pi_\theta(s, a) = \phi_{s,a} - \nabla_{\theta} \log(\sum_{a'} \exp(\theta^T \phi_{s,a'}))$$

$$\text{where } \nabla_{\theta} \log(\sum_{a'} \exp(\theta^T \phi_{s,a'})) = \frac{\sum_{a'} \exp(\theta^T \phi_{s,a'}) \phi_{s,a'}}{\sum_{a'} \exp(\theta^T \phi_{s,a'})}$$

$$= \frac{\sum_{a'} \exp(\theta^T \phi_{s,a'}) \phi_{s,a'}}{\sum_{a'} \exp(\theta^T \phi_{s,a'})}$$

$$= \sum_{a'} \pi_\theta(s, a') \phi_{s,a'} = \mathbb{E}_{a' \sim \pi(s)} [\phi(s, a')]$$

$$\nabla_{\theta} \log \pi_\theta(s, a) = \phi_{s,a} - \mathbb{E}_{a' \sim \pi(s)} [\phi(s, a')]$$

$$\text{policy gradient: } \sum_{s,a} \nabla_{\theta} \log \pi_\theta(s, a) \nabla_{\theta} \log(\pi_\theta(s, a)) A$$

$$\text{本问题因为对称性, } d^{\pi}(s_1) = d^{\pi}(s_2) = \frac{1}{2} \cdot \frac{1}{1-\gamma}$$

$$\text{有对称性, } P(s_t = s_1 | s_1, \pi) = 0$$

$$R(s_1) = \sum \gamma^t \theta^t = \frac{1}{1-\gamma}$$

$$d^{\pi}(s_2) = \frac{1}{1-\gamma} - d^{\pi}(s_1)$$

$$\text{如果 } \pi \text{ 是 conservative 的, 则 } \pi \text{ 是 uniform, } \pi \text{ 选 } a_1$$

$$\Rightarrow \pi_2(a_1), \pi_2(a_2) \Rightarrow V^{\pi_2}(s_0) = \pi_2(a_1) \cdot Q(s_0, a_1) + \pi_2(a_2) \cdot Q(s_0, a_2)$$

$$\text{直接算 } V^{\pi_2}(s_0) \Rightarrow \text{求 } V^{\pi_2}(s_0) - V^{\pi_1}(s_0)$$

$$\propto \alpha \Rightarrow \text{the degradation is proportional to } \alpha$$

$$\text{so for sufficiently small } \alpha, \text{ the policy will not flip the action.}$$

span 证明的例题

(Average reward setting)

Let v^k denote the value vector at k^{th} iteration of value iteration. In the infinite horizon average reward setting, value iteration:

$$v^k \leftarrow L v^{k-1} \quad \text{where } L v := \max_a R(s, a) + \sum_{s'} P(s, a, s') v(s')$$

it converges in the case if $\text{span}(v^k - v^{k-1}) \rightarrow 0$ where $\text{span}(v) = \max_i v_i - \min_i v_i$

(a) prove the convergence as long as, for $\alpha < 1$ $\text{span}(L v - L v') \leq \alpha \text{span}(v - v')$

$$\text{proof: } \text{span}(L v - L v') = \text{span}(L v - L v^{k-1} + L v^{k-1} - L v')$$

$$\leq \alpha \text{span}(v^k - v^{k-1}) \leq \dots \leq \alpha^k \text{span}(v - v') \rightarrow 0$$

$$\text{Bellman operator } L: L\pi: \mathbb{R}^S \rightarrow \mathbb{R}^S$$

$$[L\pi](s) := \max_{a \in A} R(s, a) + \gamma \sum_{s'} P(s, a, s') v(s')$$

$$[L^{\pi}](s) := \mathbb{E}_{a \sim \pi(s)} [R(s, a) + \gamma \sum_{s'} P(s, a, s') v^{\pi}(s')]$$

Value iteration in infinite discounted reward

1. start with any v^0 , specify $\epsilon > 0$
2. Repeat for $k = 1, 2, \dots$ until $\|v^k - v^{k-1}\|_{\infty} \leq \epsilon$
3. output a near-optimal policy $\pi(s) \in \arg \max_a R(s, a) + \gamma \sum_{s'} P(s, a, s') v^k$

Property of L-operator

$$\|L v - L u\|_{\infty} \leq \gamma \|v - u\|_{\infty}$$

$$\|L^{\pi} v - L^{\pi} u\|_{\infty} \leq \gamma \|v - u\|_{\infty}$$

$$v^* = L v^* \quad (\text{Bellman}) \quad v^{\pi} = L^{\pi} v^{\pi}$$

$$\text{proof of convergence } L^{\pi} v^k = L v^k$$

$$\|v^k - v^{k-1}\|_{\infty} = \|L v^{k-1} - L v^{k-2}\|_{\infty} \leq \gamma \|v^{k-1} - v^{k-2}\|_{\infty} \leq \dots \leq \gamma^k \|v^1 - v^0\|_{\infty}$$

$$\|v^k - v^{k-1}\|_{\infty} \leq \gamma^k \|v^1 - v^0\|_{\infty} \leq \gamma^k \cdot \frac{1}{1-\gamma} \|v^1 - v^0\|_{\infty}$$

(d) prove the span contraction property when $P(i, a, j) > 0 \forall i, j, a$

$$L v(s) - L v'(s) = (\max_a R(s, a) + P(s, a, T) v(T)) - (\max_a R(s, a) + P(s, a, T) v'(T))$$

$$\leq R(s, a) + P(s, a, T) v(T) - R(s, a) - P(s, a, T) v'(T)$$

$$= P(s, a, T) (v(T) - v'(T)) \leq (1-b) \max_j (v_j - v'_j) + b \sum_j (v_j - v'_j)$$

$$L v(s) - L v'(s) \leq (1-b) \min_j (v_j - v'_j) + b \sum_j (v_j - v'_j)$$

TD-learning

Example consider MDP with s_1, s_2 and one action $s_1 \rightarrow s_2, s_2 \rightarrow s_1, r(s_1) = 0, r(s_2) = 1$

$$V_0(s) = \theta \phi(s), \quad \phi(s_1) = (1, 0), \quad \phi(s_2) = (0, 1)$$

$$(a) \text{TD}(0) \quad \theta \leftarrow \theta + \alpha (r_t + \gamma V_{t-1} - V_t) \quad \text{where } b_t = r_t + \gamma \theta^T \phi(s_t) - V_t$$

(b) show that it can not exactly reflect true value. $V(s_1) = \gamma V(s_2) \quad V(s_2) = 1 + \gamma V(s_1)$

$$\Rightarrow V(s_2)/V(s_1) = 1/\gamma \text{ but } V(s_2)/V(s_1) = 2$$

only if $\gamma = \frac{1}{2}$, generally not representable

(c) why TD may fail to converge, and why this issue not arise in tabular settings?

the projected Bellman equations (with θ) has no solution, so TD updates can oscillate

function approx introduces shared parameters and won't converge to anything; in tabular settings, the representation is exact, Bellman operator is a contraction ensuring convergence.

policy iteration using Q-values

1. initialize π^0
2. for $k = 0, 1, 2, \dots$

$$\text{(cp)} \quad \pi^{k+1}(s) = \arg \max_a Q^k(s, a) \quad \forall s$$

TD-learning (Q-value Bellman equation)

1. initialize $Q^0 \in \mathbb{R}^{S \times A}$
2. for $k = 0, 1, 2, \dots$

$\rightarrow$ observe s_k , take action a_k ϵ -greedily

$\rightarrow$ observe r_k , next state s_{k+1}

$\rightarrow$ improve one-point Q-value as $Q^k(s_k, a_k) \leftarrow (1-\alpha) Q^k(s_k, a_k) + \alpha [r_k + \gamma \sum_{a'} \pi(s_{k+1}, a') Q^k(s_{k+1}, a')]$

π MC

simulate policy π for T steps (large T)

For $t = 1, 2, \dots, T$, observe s_t , take action $a_t \sim \pi(s_t)$, observe r_t, s_{t+1}

output trajectory $T = (s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_T)$

$$Q^{\pi}(s_t, a_t) \leftarrow r_t + \gamma r_{t+1} + \dots + \gamma^{T-t} r_T$$

Deep TD-learning

repeat for $t = 1, 2, \dots, T$

Actor-critic algorithm

in each iteration $\rightarrow$ (cp-critic) use TD-learning or MC to fit parameters w , such $f_w(s, a) \approx Q^{\pi_{\theta}}(s, a)$

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k \mathbb{E} [f_w(s, a) \nabla_{\theta} \log \pi_{\theta}(s, a)]$$

policy iteration with values

$$\text{(cp)} \quad \pi^{k+1}(s) = \arg \max_a [R(s, a) + \gamma \sum_{s'} P(s, a, s') V^k(s')]$$

HW2. T1

suppose $Q(s, a) = Q^*(s, a) + \epsilon_{s,a}$

Now, suppose for (s, a) , we have a new observation (s, a, s', r) . do Q-learning

$$Q(s, a) \leftarrow (1-\alpha) Q(s, a) + \alpha (r + \gamma \max_{a'} Q(s', a'))$$

(a) prove the updated Q will be positively biased.

By Jensen ineq. (for convex "max" func)

$$\mathbb{E} [\max_{a'} Q(s', a)] \geq \max_a \mathbb{E} [Q(s', a)] = \max_a Q^*(s', a)$$

assume at least two actions. Let a^* be optimal at $Q^*(s', a^*) = \max_a Q^*(s', a)$

because the noise non-trivial, with prob that some suboptimal action satisfies $Q^*(s', a) + \epsilon_a > Q^*(s', a^*) + \epsilon_{a^*}$

$$\therefore \mathbb{E} [\max_a Q(s', a)] > \max_a Q^*(s', a)$$

(c) for double Q-learning

$$\mathbb{E} [Q^{(2)}(s', \arg \max_a Q^{(1)}(s', a))] = \mathbb{E}_{a^{(1)}} [Q^*(s', a^{(1)})]$$

$$\leq \max_a Q^*(s', a)$$

HW1 T3

Given policy π , suppose for some (s, a) , $Q^{\pi}(s, a) > V^{\pi}(s)$

then (c) π is not an optimal policy.

because $V^{\pi}(s) = \sum \pi(s, a) Q^{\pi}(s, a)$ which is convex

Q^{π} . If there exists a with $Q > V$, then $\pi(s)$ assigns positive prob to some action with $Q^{\pi}(s, a) > V^{\pi}(s)$, hence π is not greedy w.r.t. its own Q^{π} . Define a π' that is identical to π except that plays action a deterministically in s , then $V^{\pi'}(s) = Q^{\pi}(s, a) > V^{\pi}(s)$.

HW1 T4

(a) false $v \leq L v$

(b) True. For any π , $V^{\pi} \leq L V^{\pi}$

Fix any s , $L V^{\pi}(s) = \max_a (R(s, a) + \gamma \sum_{s'} P(s, a, s') V^{\pi}(s'))$

$\geq V^{\pi}(s)$ $\forall s$.

shortest path problem. (S, A, P, R, γ) 的 Reward

$S := V(\text{nodes}) \quad A := V$ 不想让某些 action 用

P : for every edge (i, j) , on taking action j in state

$i \neq t$, transit to j w.p. 1. All other actions lead to self-loop.

All actions at t lead to self-loop.

R : For $i \neq t$, $R(i, a) = -c(i, a)$ for $\forall i, a \in E$

$R(t, a) = -c_0$ if they visit a

For $i = t$, $R(i, a) = 0$